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Reasoning

I chose VR fitness training as the mini world for the project. Members can find/select workouts, follow trainers, and work out inside a virtual world. The database keeps track of members' workouts and sessions. I utilized the FitXR website, which provides details on the trainer workout classes, virtual worlds, and workout history. I explored their website and did not have any interviews or surveys. The provided data is all fabricated. The database has members, trainers, workouts, exercises, and sessions. It stores exercises in each workout, and it displays members' workout history. Payments, subscriptions, and equipment were all ignored.

I started by breaking down the project brief and researching FitXR to decide what data should be stored in the database. I then chose the entities, attributes, and relationships by reading the slides from Chapters 3 and 4.

I then designed the Chen diagram and drew the UML diagram in MySQL Workbench. I then created the tables in SQL and linked the tables together by adding primary and foreign keys. So it has 6 tables 4 the tables logins contains the names, e-mail, and date joined, and the trainer table contains the trainer names and specialities.

The table of exercises shows the name and the type of each exercise. The exercise table has the name of each exercise. The table of workouts has the name, trainer, difficulty, length and virtual environment of the workout.

Workout_exercise table connects exercises and workouts and records the order and number of seconds in each exercise. Session table that records member, date, length, calories and score for each workout session.

One trainer can have many workouts for one workout. One member and one workout. Sessions may have the same member doing the same workout multiple times, so one session has a unique session ID. There are many exercises in each workout, and each exercise belongs to many workouts. The workoutexercise table maintains this many-to-many relationship with the exerciseid, workout_id together forming a primary key.

Another problem was telling the difference between a workout and a session. A workout is the set of exercises I plan, a session is the member doing the workout. I couldn't quite figure out where to put the sequence and length of exercises. Because one exercise might be part of different sessions, in a different order or for a different length of time in different workouts, it was wiser to include it in workout_exercise. I then added fake sample records to each table and submitted some queries to see that all the data was being held and displayed correctly.

Sample data had 4 members, 3 trainers, 4 workouts, 5 exercises, 8 workout_exercises, and 6 sessions. All 7 SELECT queries worked. MySQL refused to insert a duplicate primary key

and refused to insert a session with an invalid member ID. With it finished, this database enables us to keep track of the activities members have engaged in during their VR sessions.

Relationships between tables have been preserved so we don't need to include the data of a member, trainer, or exercise into each session.

I began by analysing the project brief and doing some research into FitXR to determine what data should be held within the database. Then I selected the entities, attributes, and relationships by using the Chapter 3 and Chapter 4 slides. I designed the Chen diagram and drew the UML diagram in MySQL Workbench. I built the tables in SQL and linked the tables using primary and foreign keys. I then populated the tables with fictional sample records, then submitted some queries to verify that the data was being held and displayed correctly.

References

FitXR. Step Into FitXR. <https://fitxr.com/pages/meta-quest>

FitXR. Your Journey: Workouts Built Around You.

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